Movement of a metal ball in a wire field

X23 (2023) — English translation

As part of this task, you will study the interaction of an uncharged metal homogeneous ball with an infinitely long thin stationary wire charged uniformly along its length. At all points of the problem, the radius R and mass m of the ball, as well as the linear charge density of the wire λ are considered known. Parts A,B and C are devoted to the study of the interaction of a ball and a wire at larger distances r between them compared to R. Parts D and E are devoted to accurately describing their interactions.

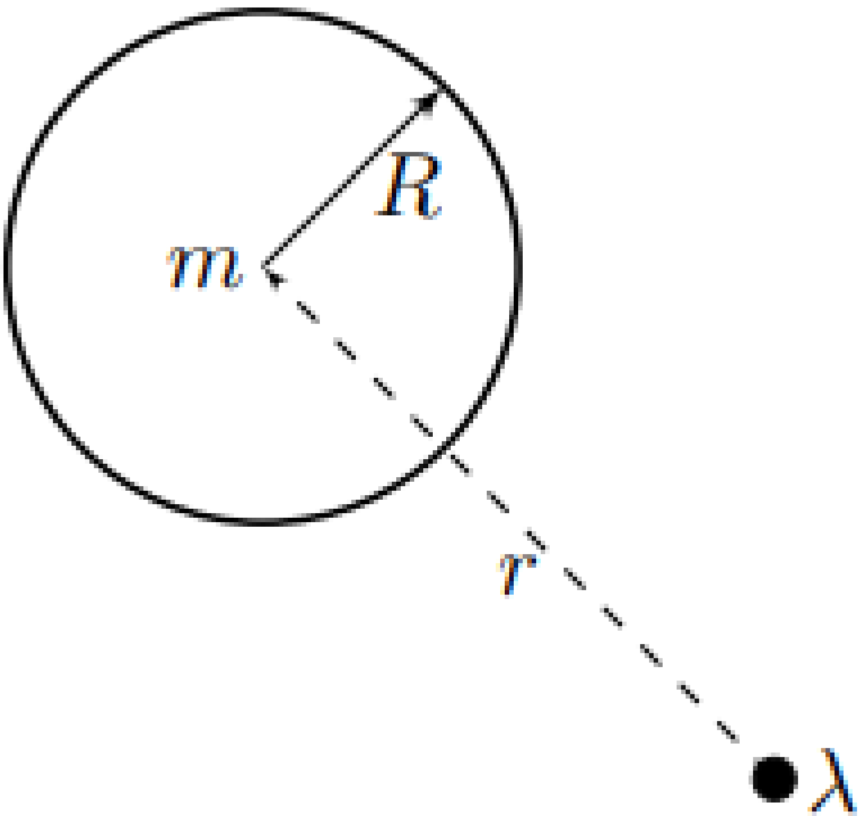

Figure 1: Figure 1.

Part A. Interaction of a ball with a wire over long distances (2.4 points)

In this part of the problem you have to obtain the dependences of the force F and potential energy W_p of the interaction between the ball and the wire. Note: At all points of parts A - C, consider that the distance r between the center of the ball and the wire is large compared to the radius R of the ball.

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Let's solve the following auxiliary problems:

A1	A metal ball of radius R is placed in a uniform electric field $\vec{E}_0$. Find the dipole moment $\vec{p}$ acquired by the ball. Express your answer in terms of $\vec{E}_0$, R and the electric constant ϵ_0.	**0.80**

A2	Let the ball be located in an electric field directed along the z axis, the intensity of which varies according to the law $$\vec{E} \approx (E_0 + kz)\,\vec{e}_z,$$ where E_0 and k are known and $kR \ll E_0$. Find the force F_z acting on the ball. Express your answer in terms of E_0, k, R, and the electric constant ϵ_0.	**0.50**

Let's move on to the interaction of the ball with the wire.

A3	Find the magnitude of the electric field of the wire E_p at a distance r from it. Express your answer in terms of λ, r and the electric constant ε_0.	**0.20**

A4	Find the magnitude and direction (attraction or repulsion) of the force F acting on the ball from the side of the wire at a distance r from it. Express your answer in terms of λ, R, r and the electric constant ε_0.	**0.50**

A5	Show that the mechanical potential energy W_p of a ball in the field of a wire at a distance r between the wire and the center of the ball is equal to $$W_p = -\frac{A}{r^2}.$$ Find A. Express your answer in terms of λ, R, and the electric constant ε_0.	**0.40**

In the future, in all points of parts B and C, use the expressions for the force and potential energy of interaction between the ball and the wire obtained in points A4 and A5, respectively. Note: You will only need the expression for the value A in point **C4**.

Part B. No contact condition (0.8 points)

The ball flies in the direction "to the wire" from infinity with an initial speed v_0 and an impact parameter b.

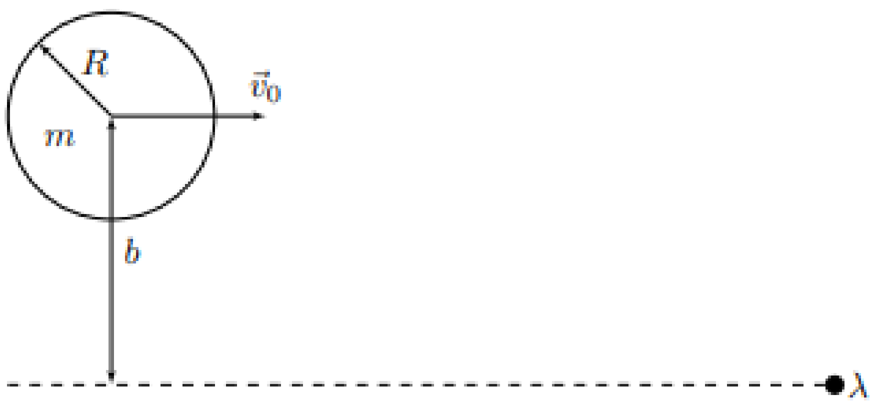

Figure 2: Figure 2.

In this part of the problem, you need to approximately obtain the conditions under which there will be no further contact between the ball and the wire.

B1	Find the minimum distance $r_{\min}$ between the center of the ball and the wire during motion, assuming that no collision occurs. Express your answer in terms of A, b, m and v_0.	**0.60**

When the initial speed v_0 of the ball is less than the critical value v_{crit}, the ball collides with the wire.

B2	Find v_{crit}. Express your answer in terms of A, b and m.	**0.20**

Part C. Trajectory (2.5 points)

To describe the trajectory of the center of the ball, we introduce a polar coordinate system (r, φ) with the origin on the wire axis. We will count the angle φ from the direction to infinity.

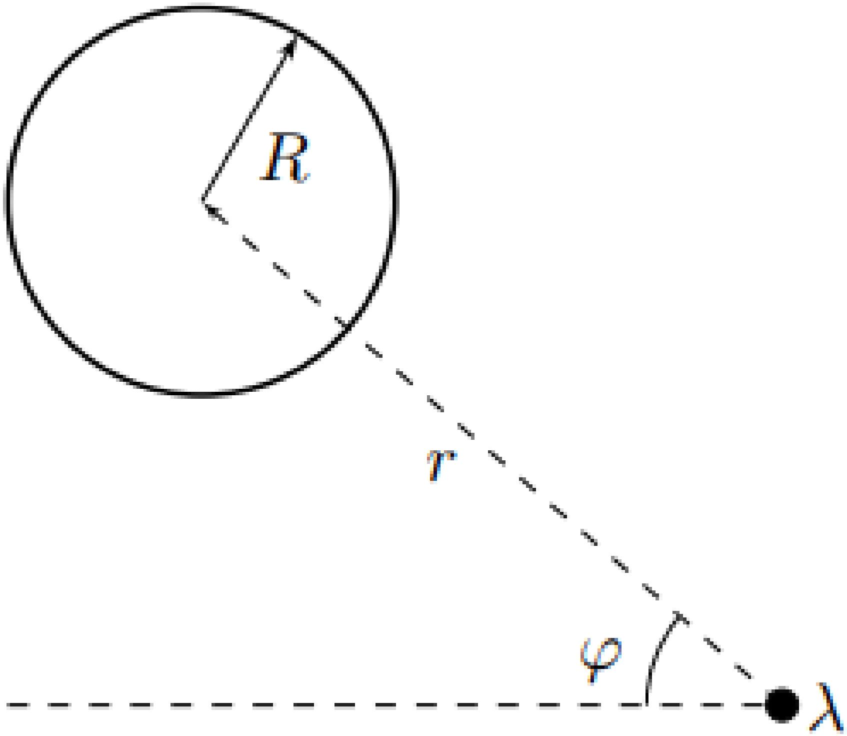

Figure 3: Figure 3.

We also use Binet's substitution:

$$u = \frac{1}{r}.$$

Denote $\dfrac{du}{d\varphi}$ by u', and $\dfrac{dr}{dt}$ and $\dfrac{d\varphi}{dt}$ by $\dot{r}$ and $\dot{\varphi}$, respectively.

C1	Express u' in terms of r, $\dot{r}$ and $\dot{\varphi}$. Find $u'(0)$ corresponding to the value of $\varphi = 0$.	**0.40**

C2	Express the mechanical energy E of the ball in terms of m, u, u', v_0, b and A.	**0.30**

C3	Get addicted $u(\varphi)$. Express your answer in terms of m, v_0, b, A and φ.	**0.70**

C4	On the answer sheet, qualitatively construct the trajectory of the center of the ball in coordinates u,φ. The coordinates of all feature points can be expressed in terms of m, v_0, b and A.	**0.60**

C5	Using the value of the parameter A found in step **A4**, find the initial speed of the ball v_1 at which during the interaction between it and the wire the direction of the ball's speed changes to the opposite. Express your answer in terms of m, b, R, λ and the electrical constant ϵ_0.	**0.50**

The equations obtained in the first three parts of the problem are not always applicable, but this problem can be solved exactly. The question of the trajectory will not be studied further, but the conditions for the absence of contact between the ball and the wire will be precisely obtained.

Part D. Finding the force of interaction between a ball and a wire using the image method (3.5 points)

Let's solve the auxiliary problem again. Consider a point charge q_1 located at a distance L from the center of an uncharged metal ball. As is known, this problem is solved by the image method: in the center of the ball and at a certain distance $S<R$ it is necessary to place charges q_2 and $-q_2$, respectively, ensuring a constant potential on the surface of the ball.

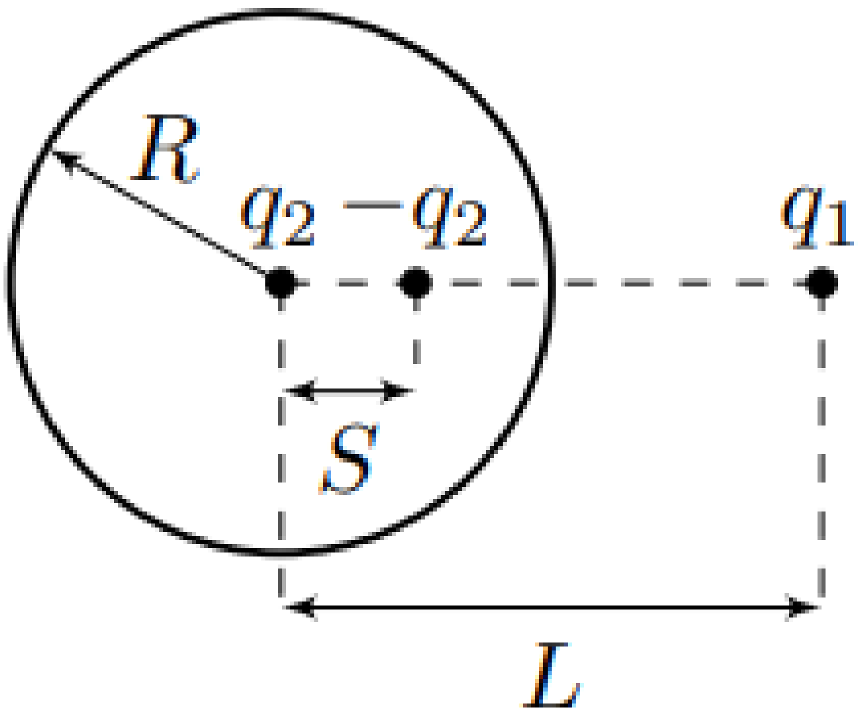

Figure 4: Figure 4.

Note: You may need the following antiderivative:

$$\int \frac{dx}{a^2+b^2x^2} = \frac{1}{ab}\arctan\left(\frac{bx}{a}\right) + C.$$

D1	Find S and q_2. Express your answers using R, L and q.	**0.60**

Let's move on to the interaction of the ball with the wire. The center of an uncharged metal ball of radius R is at a distance r from the wire.

Let us draw two lines from the center of the ball, constituting angles α and $\alpha + d\alpha$ with the direction "to the wire". We denote the charge of the wire element in the area of intersection with the lines as dq_1, the charge-images of this element as dq_2 and $-dq_2$, and the distances from the charge $-dq_2$ to the center of the ball and to the wire as S and r_-, respectively.

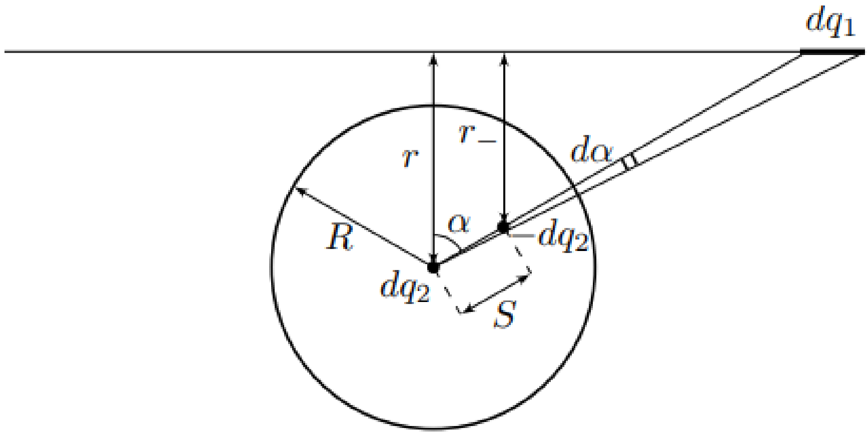

Figure 5: Figure 5.

D2	Find dq_1, dq_2, S and r_-. Express your answers using λ, R, r, α and $d\alpha$.	**0.40**

D3	Find the magnitude and direction (attraction or repulsion) of the vector sum of forces $d\vec{F} = d\vec{F}_+ + d\vec{F}_-$ interaction of charges dq_2 and $-dq_2$ with the wire. Express your answer in terms of λ, R, r, α, $d\alpha$ and the electrical constant ϵ_0.	**0.20**

To describe the force and potential energy of interaction between the ball and the wire, it is convenient to introduce the angle θ, defined as follows (see figure):

$$\theta = \arcsin\left(\frac{R}{r}\right) = \arctan\left(\frac{R}{\sqrt{r^2 - R^2}}\right).$$

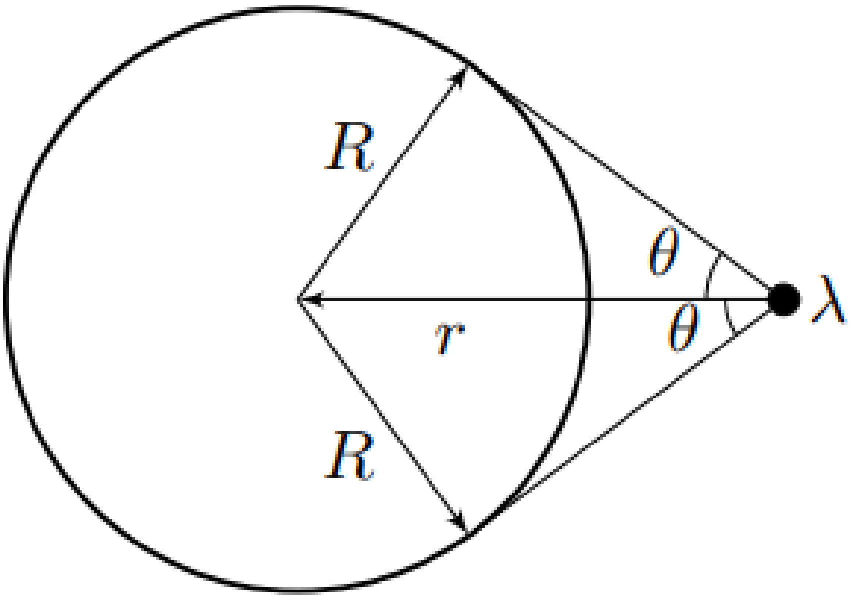

Figure 6: Figure 6.

D4	Find the magnitude and direction of the total force of interaction between the ball and the wire F_{full}. Express your answer in terms of λ, R, θ and the electric constant ϵ_0.	**1.00**

D5	Find the potential energy W_p of the interaction of the ball with the wire. Express your answer in terms of λ, R, θ and the electric constant ε_0.	**1.00**

D6	What speed v will an uncharged metal ball of radius R and mass m have when colliding with a wire if it is released from infinity without an initial speed? Express your answer in terms of λ, R, m and the electric constant ε_0.	**0.30**

Part E: Precise No-Contact Conditions (2.8 points)

An uncharged metal ball of radius R and mass m flies from infinity onto a wire with a speed v_0 and an impact parameter $b > R$. In this part of the problem, for different relations b/R we will obtain restrictions on the speed v_0 at which the ball moves without contacting the wire. The minimum possible speed at which the ball moves without contact with the wire will be denoted by v_{crit}. Let us denote the parameter θ at the moment of maximum approach of the ball to the wire as θ_{max}.

E1	For a given value v_0, obtain the dependence of the square of the radial velocity component $\dot{r}^2$ on the parameter θ. Express your answer in terms of m, v_0, λ, R, b, electric constant ε_0 and θ.	**0.40**

E2	Construct high-quality graphs of the dependence $\dot{r}^2(\theta)$ for different values of v_0.	**0.60**

E3	Considering all quantities except v_0 constant, you obtain a system of equations that allows you to determine the value v_{crit}. The system of equations can include m, R, b, λ, electric constant ε_0, θ_{max} and v_{crit}.	**0.60**

E4	Find v_{crit1}, v_{crit2} and v_{crit3} corresponding to the following values of b: $$b_1 = 20R, \qquad b_2 = 3R, \qquad b_3 = 1.1R.$$ Express your answers in terms of m, R, λ, and the electric constant ε_0.	**1.20**

Source: `https://pho.rs/p/2902`